Lesson 1**

LOGARITMIC DIFFERENTIATION

(*) One of the classical method of differentiation is the logarithmic differentiation. It requires only 2 steps:

Given $$y = \frac{f(x)\, g(x)^{d(x)}}{h(x)}$$

Step 1: (Take log or $\ln(\cdot)$ both sides)

$$\ln y = \ln\left[\frac{f(x)\, g(x)^{d(x)}}{h(x)}\right] = \ln[f(x)] + d(x)\ln[g(x)] - \ln[h(x)]$$

Step 2: (differentiate)

$$\frac{1}{y}\frac{dy}{dx} = \frac{f'(x)}{f(x)} + d'(x)\ln[g(x)] + d(x)\cdot\frac{g'(x)}{g(x)} - \frac{h'(x)}{h(x)}$$

$$\frac{dy}{dx} = y\cdot\left[\frac{f'(x)}{f(x)} + d'(x)\ln[g(x)] + d(x)\frac{g'(x)}{g(x)} - \frac{h'(x)}{h(x)}\right]$$

Hence $$\frac{dy}{dx} = \frac{f(x)\, g(x)^{dx}}{h(x)}\left[\frac{f'(x)}{f(x)} + d'(x)\ln[g(x)] + d(x)\frac{g'(x)}{g(x)} - \frac{h'(x)}{h(x)}\right]$$

Example 1

Differentiate the following:

(a) $y = \sqrt{3x}^{\sqrt{x}}$ (b) $y = (x+2)^3(x-5)^4$

Solution

Step 1: $\ln y = \sqrt{x}\,\ln(\sqrt{3x})$

Step 2: $\frac{1}{y}\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\ln(\sqrt{3x}) + \sqrt{x}\cdot\left(\frac{3}{\sqrt{3x}}\right) = \frac{1}{2\sqrt{x}}\ln(\sqrt{3x}) + \sqrt{3}$

$$\therefore \frac{dy}{dx} = y\left[\frac{1}{2\sqrt{x}}\ln(\sqrt{3x}) + \sqrt{3}\right] = \sqrt{3x}^{\sqrt{x}}\left(\frac{1}{2\sqrt{x}}\ln(\sqrt{3x}) + \sqrt{3}\right)$$

(b) $y = (x+2)^3(x-5)^4$

$$\ln y = 3\ln(x+2) + 4\ln(x-5)$$

$$\frac{1}{y}\frac{dy}{dx} = 3\cdot\left(\frac{1}{x+2}\right) + 4\left(\frac{1}{x-5}\right)$$

$$\frac{dy}{dx} = y\left[\frac{3}{x+2} + \frac{4}{x-5}\right]$$

$$\frac{dy}{dx} = (x+2)^3(x-5)^4\left[\frac{3}{x+2} + \frac{4}{x-5}\right]$$

$$= (x+2)^3(x-5)^4\left[\frac{3(x-5)+4(x+2)}{(x+2)(x-5)}\right]$$

$$= (x+2)^3(x-5)^4\left[\frac{7x-7}{(x+2)(x-5)}\right]$$

$$\therefore \frac{dy}{dx} = 7(x+2)^2(x-5)^3(x-1)$$

Example 2

Differentiate the following:

(a) $y = \dfrac{e^{2x}\cos 3x}{\sqrt{5-3x^2}}$ (b) $y = (\ln x)^x$, $x>0$

Solution

$$\ln y = 2x\ln e + \ln(\cos 3x) - \ln(5-3x^2)^{1/2}$$ Since $\ln e = 1$

$$\ln y = 2x + \ln(\cos 3x) - \frac{1}{2}\ln(5-3x^2)$$

$$\frac{1}{y}\frac{dy}{dx} = 2 + \frac{-3\sin 3x}{\cos 3x} - \frac{1}{2}\cdot\left(\frac{-6x}{5-3x^2}\right)$$

$$\frac{dy}{dx} = \frac{e^{2x}\cos 3x}{\sqrt{5-3x^2}}\left[2 - 3\tan 3x + \frac{3x}{5-3x^2}\right]$$

(b) $y = (\ln x)^x$

$$\ln y = x \ln[\ln x]$$

$$\frac{1}{y}\frac{dy}{dx} = \ln[\ln x] + x \cdot \left(\frac{1/x}{\ln x}\right)$$

$$\frac{dy}{dx} = y\left[\ln(\ln x) + \frac{1}{\ln x}\right]$$

$$\therefore \frac{dy}{dx} = (\ln x)^x\left[\ln(\ln x) + \frac{1}{\ln x}\right]$$